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Comparison of semi-empirical models, symbolic regression, and machine learning approaches for prediction of tensile strength in steels

Gerfried Millner [a], Gabriel Kronberger [c], Manfred Mücke [a], Lorenz Romaner [b], Daniel Scheiber [a,*]

[a] Materials Center Leoben Forschung GmbH, Roseggerstrasse 12, Leoben, 8700, Austria
[b] Department of Materials Science, Montanuniversität, Leoben 8700, Austria
[c] University of Applied Sciences Upper Austria, School of Informatics, Communications and Media, Softwarepark 11, Hagenberg, 4232, Austria

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ABSTRACT

We employ data-driven models to predict the tensile strength of steel coils using information on their chemical composition and process parameters. The dataset contains extensive chemical analyses, diverse process parameters, and the characterized tensile strength as target property. We compare prediction quality and traceability of the predictions of pure machine learning models and physics-informed models. To introduce physical knowledge, we combine models from literature knowledge with symbolic regression and compare the physics-inspired models to machine learning models. In contrast to classic black-box models, symbolic regression provides mathematical equations for the estimation of the target value, facilitating straightforward interpretation. To analyze the predictions from classic black-box machine learning models, we use feature importance analysis with SHAP and contrast the obtained feature impacts with physics-based model parameters. We find that for the present use case, Artificial Neural Networks perform best, while the physics-infused models from symbolic regression allow for better interpretability.

1. Introduction

Steel production is responsible for about 11% of global CO2 emissions, according to the Steel Climate Impact report (Hasanbeigi, 2022). The need to reduce this contribution is immense and one way to achieve this reduction is by an increased use of scrap steel instead of iron ore. As shown in a recent report (Fraunhofer Institute for Environmental, Safety, and Energy Technology (UMSICHT), BDSV (Bundesverband der Deutschen Stahlindustrie), 2022), the CO2 emissions can be significantly reduced by the use of steel scrap in Europe and could lower the costs of climate change by up to 20 billion euros a year. However, the usage of scrap introduces many unwanted elements, which influence the microstructure evolution during processing and, therefore, mechanical properties in a way not fully understood yet. Modeling steel production and the impact of chemical composition is necessary to understand and counteract the influence of alloying and tramp elements.

The effect of key elements like niobium (Nb), manganese (Mn), chromium (Cr), and titanium (Ti) on the process–structure–property (PSP) relationships in steel has been extensively discussed in literature, see e.g. Refs. Hudd (2006) and Hutchinson (1984). Additionally, significant attention has been devoted to the role of aluminium (Al) and nitrogen (N), leading to AlN precipitation,

* Corresponding author.
E-mail addresses: gerfried.millner@mcl.at (G. Millner), daniel.scheiber@mcl.at (D. Scheiber).

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affecting both steel microstructure and mechanical properties, e.g. Refs. Kozeschnik, Pletenev, Zolotorevsky, and Buchmayr (1999), Radis and Kozeschnik (2010a, 2010b), Wilson and Gladman (1988) and Zolotorevsky, Pletenev, and Titovets (1998). More recently, the effect of impurity elements on the microstructure and mechanical properties of a low carbon steel has been investigated by Duan, Farrugia, Davis, and Li (2022) by intentionally adding copper (Cu), nickel (Ni), tin (Sn), and chromium (Cr) to mimic the occurrence of those elements commonly found in scrap materials. They reported higher strength and lower ductility with increasing impurity level.

Miranda et al. (2019) studied the effects of less common residual elements from scrap such as tungsten (W), molybdenum (Mo), and nickel (Ni), highlighting the significant role these elements can play in determining the mechanical properties of steel: While W leads to an increased hardness due to the formation of carbides (Lassner, 1999), Ni can enhance steel toughness at small quantities (Nakajima et al., 2014) and higher Mo content has been reported to increase yield and tensile strength (Junhua et al., 2004). Though the knowledge on individual solute effects on mechanical properties gives already insights into individual PSP relationships, the complete range of elements is too complex to capture with such approaches due to the manifold possible interactions between the solutes.

The modeling of mechanical parameters of steels has a long history and ranges from empirical models, to physics-based approaches, and purely data driven methods (Hudd, 2006; Hutchinson, 1984). Modeling solid solution hardening, among other things, has been investigated by Sieurin et al. leading to linear models of mechanical properties such as tensile strength and yield stress from chemical composition, grain size, ferrite content and product thickness (Sieurin, Zander, & Sandström, 2006). Those models include several fitting parameters that need to be adjusted for a specific steel, in order to get satisfactory results.

Machine Learning (ML) methods have been increasingly employed for predicting mechanical properties of alloys in materials science, as reviewed by e.g. Durodola (2022). The successful application of data driven approaches in steel production has been shown by many authors, e.g. Xie et al. (2021) predicting yield stress, tensile strength, elongation, and impact energy online in a real steel plant. Millner, Mücke, Romaner, and Scheiber (2023) employed many different ML models to predict tensile strength, yield stress, elongation at fracture, and r-value using data originating from the manufacturing of cold-rolled, batch-annealed low-carbon steel coils. Although other models can achieve comparable prediction quality, artificial neural networks performed the best using this data. Further noteworthy examples include the work of Guo, Yu, Liu, Wang, and Jiang (2019), Wu et al. (2020) and Xu, Liu, and Zhang (2019) investigating the effect of chemical composition along with some process parameters, on tensile strength, yield stress and elongation. While classical ML methods have been shown to be of great help to interpolate for materials and processes where extensive data is available, they are of limited help to improve the general knowledge on the underlying mechanisms like solid solution hardening or to transfer models from one material or process to another material or process as it is typically very difficult to extract the relations learned by such black-box models.

To improve explainability of black-box models, Symbolic Regression (SR) is a viable method (Kronberger, Burlacu, Kommenda, Winkler, & Affenzeller, 2024). The advantage of SR compared to most other machine learning methods is the fact that it expresses the input–output relation by an analytic expression (using a specified set of arithmetic operators). This allows manual investigation of the found relations and guiding search by constraining the operator set as well as the expression structure. See Wang, Wagner, and Rondinelli (2019) for applications of the method in material science. Piringer, Bloder, and Kronberger (2022) used SR for modeling of steel phase kinetics in steel production. Xiong, Zhang, and Shi (2020) applied SR to generate analytic expressions for tensile strength and other steel properties and show the great potential of this approach.

In this work, we are focusing on the prediction of tensile strength of steel coils with various data-driven methods. The studied methods include a semi-empiric model from literature which describes the effect of solid solution hardening, Hall–Petch strengthening, and the influence of thickness on tensile strength, where we are optimizing the fitting parameters with our steel data. Furthermore, different variants of symbolic regression are applied to find mathematical expressions that connect chemical composition and process parameters to tensile strength with various degrees of physicality. Finally, we train an artificial neural network to predict tensile strength, which showed the best performance among a set of machine learning models on the same dataset in previous works (Millner et al., 2023; Millner, Mücke, Romaner, & Scheiber, 2024). We compare predictive performance, interpretability and traceability of those different approaches. This research extends beyond existing literature (Piringer et al., 2022; Xiong et al., 2020) by not only demonstrating the efficacy of data-driven methods for predicting mechanical properties of steel, but also conducting a thorough comparative analysis of black-box and non-black-box methodologies. This approach reveals connections between chemistry and tensile strength across various models, providing insights into the underlying relationships within the data.

The paper is structured as follows: In Section 2 we describe the dataset from an industrial steel coil production and necessary steps needed to transform it to a machine-readable dataset suitable for training of the various models. The models together with the respective training strategies are explained in detail in Section 3. We present results and discuss them in Section 4 and subsequently highlight the impact of chemistry in all our models. Section 5 concludes.

2. Data acquisition and pre-processing

In this chapter, the origin of the used dataset (Section 2.1) and all steps to create the input data for our models (Section 2.2) are presented.

2.1. Dataset

The data used in this work was provided by voestalpine Stahl GmbH and consists out of a complete chemical analysis and hundreds of parameters measured throughout all stages of manufacturing cold-rolled, batch-annealed low-carbon steel coils in 22,802 observations. Upon completion of the production process, the tensile strength is measured through a tensile test on a specimen, in accordance with DIN EN ISO 6892-1. The details of the dataset and the originating process have been described previously by Millner et al. (2023).

In this study, we are selecting specific features based on the models used by Hodgson and Gibbs (1992) and Sieurin et al. (2006), described in more detail in Section 3.1.1.

The 19 raw features comprise the weight percent of 14 chemical elements (Al, C, Cr, Cu, Mn, Mo, N, Ni, P, S, Si, Sn, Ti, W), the annealing temperature during batch annealing (between 651 and 716 °C), the sheet thickness (between 0.4 and 3 mm), the holding time during batch annealing (between 275 and 23,320 min), and two length measurements to calculate the induced strain during cold rolling. Among the chemical elements, Mn with between 0.153 and 0.429 wt%, Al with between 0.015 and 0.090 wt% and C with between 0.008 and 0.129 wt% are the most abundant. All other elements have a mean concentration of below 0.04 wt%, respectively.

2.2. Data pre-processing steps

The first step in our data pre-processing process is data cleaning, which ensures the quality and reliability of the input data and is crucial for the model performance and accuracy. In this step, we exclude observations where necessary information is missing or inconsistencies due to processing and tensile test related criteria in accordance with process experts. Next, we calculate the induced true strain during cold rolling from length measurements before (L_0) and after (L) this process step with $\epsilon = \log(1+(\frac{L-L_0}{L_0}))$ (Callister, 2007), resulting in strain values between 0.65 and 1.69. This results in a slightly reduced dataset with now only 18 features and 16,649 observations.

3. Methodology

We compare four different models predicting tensile strength regarding prediction quality and interpretability. Specifically:

- SHM: We apply a model based on semi-empirical findings from the literature (Hodgson & Gibbs, 1992; Sieurin et al., 2006), describing the effects of solid solution hardening, Hall–Petch strengthening, and the influence of thickness and name it Sieurin–Hodgson model (SHM).
- AHM: Then, we utilize the physical knowledge gained from this model to construct analytical models similar to the SHM using symbolic regression, resulting in the analytical hybrid model (AHM).
- ABBM: We allow more freedom to find an analytical expression in an analytical black-box model (ABBM). Although symbolic regression generates an expression that may allow physical interpretation, we here treat the symbolic regression models simply as a black-box and allow larger and more complicated expressions.
- ANN: We apply a classic machine learning model in the form of an artificial neural network (ANN).

The model types are presented in detail in Section 3.1. Implementation and training is described in Section 3.2. The results are provided in Section 4, accompanied by a comprehensive analysis of the interpretability of the models. Interpretability is investigated via manual inspection of expression for SHM and AHM, and via SHAP for ABBM and ANN.

3.1. Model types

3.1.1. Sieurin–Hodgson model (SHM)

For the semi-empirical approach, we adapt the model from Sieurin et al. (2006) and fit the parameters to our dataset, because it gives a relatively good prediction quality for tensile strength ($R^2 = 0.99$) and has low complexity. The original equation from Sieurin et al. (Eq. (11) in Sieurin et al. (2006)) consists of a linear combination of chemical composition, thickness, ferrite fraction, and grain size. The model is given by

$$R_m = K + \sum_i A_{c_i} c_i^{2/3} + A_N c_N^{1/2} + A_t t^{-1/2} + f(\delta)\delta + A_g(N) d^{-1/2}, \tag{1}$$

with R_m as tensile strength, K as some constant, c as chemical concentration in weight percent, t as the sample thickness and A_i as fitting parameters, where $A_g(N)$ denotes a linear nitrogen dependence. For application to our data, it has to be modified for the use in our work: Since we have a purely ferritic steel, we are neglecting the term for the ferrite fraction ($f(\delta)$). As we do not have any information about the grain sizes (d) in our steel coils available, we combine the equation with another model from Hodgson and Gibbs (1992) to predict the average grain size:

$$d^m = \left(\epsilon^{-\frac{1}{2}}\, d_0^{0.4} \exp\left(\frac{Q_{\text{rex}}}{RT}\right)\right)^m + k\, t_{\text{growth}} \exp\left(\frac{Q_g}{RT}\right) \tag{2}$$

with ϵ as the true strain induced by cold rolling, d_0 as the initial grain size before recrystallization and grain growth in μm, R = 8.314 J/(mol K) as the gas constant, T as the holding temperature during annealing in Kelvin and t_{growth} as the time after complete recrystallization in seconds. Q_{rex} and Q_g are apparent activation energies for recrystallization and grain growth and k and m as material dependent constants.

The combination with the Sieurin equation is given by

$$R_m = K + \underbrace{\sum_i A_{c_i} c_i^{\frac{2}{3}} + A_N c_N^{\frac{1}{2}}}_{Solid\ Sol.\ Strength.(S1)} + \underbrace{A_t t^{-\frac{1}{2}}}_{(S2)} + \underbrace{A_g c_N \left[\left(\epsilon^{-\frac{1}{2}} d_0^{0.4} \exp\left(\frac{Q_{\text{rex}}}{RT}\right)\right)^m + k\, t_{\text{hold}} \exp\left(\frac{Q_g}{RT}\right)\right]^{-\frac{1}{2m}}}_{Hall-Petch\ Term(S3)} \tag{3}$$

Consisting out of a bias term (K), a solid solution strengthening term (S1), a term taking the sample thickness into consideration (S2) and a term describing the Hall–Petch strengthening (S3). As there are no per-sample measurements available for grain size, we have determined an average grain size of 10 μm from selected samples and assigned a fixed value of d_0 as 10 μm for all samples. We are approximating the time after complete recrystallization available for grain growth (t_{growth}) with the holding time during annealing (t_{hold}) and with that, we are neglecting that the recrystallization already is over during heating of the steel coils. This approximation is still reasonable, because the holding time should be proportional to the total time available for grain growth and is multiplied by a constant value (k).

To sum up, there are 21 fitting parameters in Eq. (3): one for each chemical element (in total 14), one each associated with thickness and grain size, two activation energies of recrystallization and grain growth(Q_{rex} and Q_g), k and m are associated with the grain growth term, and K is a bias term, representing further influences.

3.1.2. Analytical hybrid model (AHM)

To create analytical expressions of the strengthening process, we apply genetic programming (GP) for symbolic regression (SR). Thereby we include physical knowledge by using Eq. (1) as a structure template (Asadzadeh, Gänser, & Mücke, 2021; Kronberger et al., 2024).

The structure template for SR based on Eq. (1) reads

$$R_m = K + \sum_i A_i c_i^{2/3} + A_N c_N^{1/2} + A_t t^{-1/2} + A_g c_N f_g(R, T, \epsilon, t_{hold})^{-1/2} \tag{4}$$

whereby we again ignore the ferrite fraction term. The c_i denote the weight-percentage per chemical component (except nitrogen), c_N is the nitrogen content, t the thickness, and f_g is allowed to depend on the gas constant R. Further parameters are the holding temperature during annealing (T), the strain induced by cold rolling (ϵ) and holding time during annealing (t_{hold}). Parts that are to be identified automatically by GP include the intercept K, all coefficients A_i, A_N, A_t, A_g, and most importantly the function f_g for the average grain size which may depend on R, T, ϵ, and t_{hold} as used by Hodgson and Gibbs (1992). Additionally, we pre-processed the chemical composition before providing it as input data for the symbolic regressions search by raising all elements to the power of 2/3, except for nitrogen, which was raised to the power of 1/2, in accordance to Sieurin et al. (2006) (see Eq. (1)). By configuring the grammar for the expression trees, we make sure that only addition, subtraction, multiplication and exponential functions in the form $\exp(\frac{\cdot}{RT})$ are allowed in the expression for f_g. Our approach is similar to the approach of Asadzadeh et al. (2021) who used SR with structure templates for semi-parametric modeling of a bending process. By constraining possible SR solutions in this way, we improve the comparability with the semi-empiric approach highlighted above and introduce physical knowledge in the form of Arrhenius-like temperature influences, $\exp(\frac{\cdot}{RT})$.

3.1.3. Analytical black-box model (ABBM)

In addition to the other models described previously, we apply GP for symbolic regression (SR) without using a template and allow GP to freely evolve a model structure that fits the observations.

3.1.4. Artificial neural network (ANN)

ANNs consist of multiple layers with numerous fully connected neurons. These neurons possess weighted contributions stored in a weight matrix, which are aggregated and activated by an activation function to compute predictions (Grossi & Buscema, 2008). ANNs have been applied with great success to similar problems, e.g. Orta, Kayabasi, and Senol (2020) and Xie et al. (2021). Our ANN is structured as a feed-forward neural network, featuring a flexible number of hidden layers and nodes within them. The hyperparameter tuning was done with the hyperparameter optimization tool Optuna (Akiba, Sano, Yanase, Ohta, & Koyama, 2019). After each train/test split, an optimal set of hyperparameters is automatically determined by the best result of a 5-fold cross-validation of the training data in 100 trials. To increase the performance of the ANN, we normalize the data to zero mean and unit variance, by subtracting the population mean from each value and then dividing by the population standard deviation. Note that this is done after splitting the data to avoid data snooping. This is solely done here, as the parameters of the ANN are not significantly relevant for interpretability and hence the scale of the input does not matter.

Table 1
Comparison of the material dependent and fitting parameters in the models following Eq. (3). The minimum and maximum values of the optimized parameters obtained from five distinct cross-validation runs are presented, alongside parameters reported in literature from Hodgson and Gibbs (1992) and Sieurin et al. (2006) (Sieurin1).

Parameter	Sieurin1	SHMs: Min	SHMs: Max
K	–	180.3	182.2
A_{Al}	–	59.04	65.93
A_{C}	–	209.34	218.87
A_{Cr}	76	21.18	27.15
A_{Cu}	–	2.66	12.52
A_{Mn}	–	48.13	52.34
A_{Mo}	17	186.71	199.74
A_{N}	146	301.06	312.14
A_{Ni}	−13	25.41	33.21
A_{P}	–	299.32	307.84
A_{S}	–	−26.64	−19.15
A_{Si}	–	103.37	109.44
A_{Sn}	–	−35.26	−19.79
A_{Ti}	–	402.87	408.77
A_{W}	–	0.86	5.62
A_t	7.3	−0.04	−0.03
A_g	0.2	1.38	4.6
k	1.45E+27	1.45E+27	1.45E+27
m	7	4.33	4.76
Q_g	−4.00E+05	−7.60E+05	−6.60E+05
Q_{rex}	−4.50E+04	−7.94E+04	−6.29E+04

3.2. Model implementation and training

3.2.1. Training of SHMs

We optimized the fitting and material dependent parameters (A, Q_{rex}, Q_g, k and m) of the SHM with a least-squares solver for non-linear problems from scipy (Virtanen et al., 2020) (`scipy.optimize.least_squares`). For that purpose, we split the dataset with 5-fold cross-validation into training and testing data, fitting the parameter with the training dataset and evaluating the fitted model with the testing data. This approach leads to 5 separate models (SH (cv1-5)) and Table 1 states the range of model parameters across these five models. We additionally applied Eq. (3) with parameters used by Hodgson and Gibbs (1992) and Sieurin et al. (2006) in their work. Since many chemical elements are present in our steel and only N, Cr, Ni and Mo are considered by Sieurin, we have to set many of the corresponding coefficients to zero. Not surprisingly, this tensile strength prediction was a failure with MAE of 69.7 MPa and a negative coefficient of determination ($R^2 = -76.8$), meaning the prediction is far worse than to just guess the mean value of all observations.

3.2.2. Implementation and training of AHM

We use the software package HeuristicLab[1] (Wagner & Affenzeller, 2005), which allows to use such structure templates for symbolic regression, described in Section 3.1.2. In this SR configuration, we use gradient-based local optimization for fitting all coefficients. Each time the fitness of an individual is evaluated, we run ten iterations of a nonlinear least squares algorithm (Levenberg–Marquardt) to try to improve the coefficient values as described by Kommenda, Burlacu, Kronberger, and Affenzeller (2020), including the integration of linear scaling. If the coefficient vector is improved by the optimization procedure, the coefficients in the individual are updated, which allows solution candidates to inherit the improved coefficients in subsequent iterations of the evolutionary algorithm.

The GP algorithm returns the model with the best fitness as the solution. Since GP is a nondeterministic algorithm we execute 30 independent repetitions with different random seeds and select the solution with the best MAE on the training set for the detailed analysis. For better comparability with the results of the other models, we split the data via 5-fold cross-validation and performed the 30 runs within each fold.

3.2.3. Implementation and training of ABBM

For comparison we also use GP without structure templates to produce an alternative SR model, the ABBM. This allows us to check whether the structure template improves the prediction error, or whether the imposed structure is too rigid and has a detrimental effect on the prediction error. For these runs, we use the same algorithm parameters but a larger set of function symbols and larger trees to allow to find structurally similar models as with the structure template. The GP parameter values used in both experiments are shown in Table 2. Optimization and algorithm execution is done the same way as in the case of the AHM.

[1] https://github.com/heal-research/HeuristicLab.

Table 2
Algorithm parameters used for the SR experiments. Size and depth constraints as well as the allowed symbols in the column (AHM) refer only to the sub-expression for f_g.

Parameter	Value (AHM)	Value (ABBM)
Population size	1000	1000
Generations	200	200
Parameter optimization iterations	10	10
Max. size	50	100
Max. depth	12	15
Allowed symbols	$+, -, \times, \exp(\frac{\cdot}{RT})$	$+, \times, \div, \exp, \log, x^2, \sqrt{x}$
Mutation probability	10%	10%
Selection	Tournament (group size 4)	Tournament (group size 4)

3.2.4. Training of ANN

The ANN is implemented in python 3.11.5 using Tensorflow 2.12.0 (Abadi et al., 2015). To optimize the network, we employed the Adam optimizer (Kingma & Ba, 2014) with an initial learning rate that decreases by 10% with each epoch until reaching a minimum learning rate (determined by a learning rate factor) or the completion of training. Additionally, we integrated early stopping, terminating training when the loss converges without improvement for 20 epochs. For regularization, we used L2 regularization, treating the regularization strength as an additional hyperparameter. The hyperparameter tuning was done within the following boundaries:

- Batch Size: [1,1000]
- Learning rate (start): $[10^{-10}, 0.1]$
- Learning rate (factor) (x * learning rate (start)): [0,1]
- Number of hidden layers: [1,10]
- Number of nodes per hidden layer (18 input nodes): [18, 36, 54, 72, 90, 108, 126, 144, 162, 180]
- L2 regularization strength: $[10^{-10}, 0.1]$

The resulting hyperparameter values are given in Table 7, leading to the models ANN0 to ANN4. What stands out in the results is the very low learning rate factor found for ANN0 by Optuna, leading to a low minimum learning rate.

4. Results and discussion

In this chapter, the results of our investigation are outlined and discussed. First we are presenting results from the application of the SHM, as derived in the previous chapter in Section 4.1. Next, results from our analytical hybrid and black-box models are discussed in Sections 4.2 and 4.3. Then, we are comparing the results of those models to the predictive accuracy of a machine learning model using the same data input in Section 4.4. Finally, we are highlighting feature importance determination of all our models in Section 4.5 and investigate the influence of various chemical elements within our models.

4.1. SHM

Optimizing the parameters of the SHM structure to our dataset results in a mean R^2 value of 0.565 and a mean MAE of 4.94 MPa within a 5-fold cross-validation, with five models named SHM_0 to SHM_4 with the respective results given in Table 3.

The average impact of the respective terms is plotted in a barchart in Fig. 1. Many of the parameters differ from the ones in literature (Eq. 11 in Sieurin et al. (2006), named Sieurin1), when optimized with our data. The optimized activation energies are nearly twice as high, resulting in a stronger effect of the Hall–Petch term (S3 in (3)). Note that the exponential influence of the activation energies is balanced by the power function. Furthermore, the solid solution strengthening term has an even larger impact, see Fig. 1. The extremely high parameters of some chemical elements can be partly explained by their limited amount in the dataset, e.g. Nb and Zr. While the thickness term has nearly no influence after optimizing the parameters, since A_t has values just below zero, it has a significant impact in the model with parameters from literature. Note that the bias term K is missing in Sieurin1, while it contributes the most in the models with optimized parameters. Additionally to Sieurin1, we also calculated the results when applying Eq. 13 from Sieurin et al. (2006) (Sieurin2) to predict the tensile strength. While the parameters can be compared directly for Sieurin1, in Sieurin2 the variables for chemical composition have no exponent factor and the parameters are differing as a consequence. Sieurin is recommending K = 172 for Sieurin2, which does suit to the optimized ones but with a similar strong influence of the thickness term, leading to an overestimated prediction ($R^2 = -132$, MAE = 95.4 MPa). The discrepancies observed can be explained by the different steel grades studied: Sieurin focused on austenitic steels, while our dataset contains only ferritic steels. This fundamental difference requires comprehensive parameter refitting to ensure accurate predictions and leads consequently to deviating parameters.

Table 3
Comparison of the prediction quality of literature models applied with our data (Sieurin1 and Sieurin2) and of Eq. (3) after least squares optimization of parameters using a 5-fold cross-validation (SHM (mean) and SHM_{0-4}).

Metric	Sieurin1	Sieurin2	SHM (mean)	SHM_0	SHM_1	SHM_2	SHM_3	SHM_4
R^2	−76.8	−132	0.565	0.552	0.574	0.589	0.557	0.551
MAE [MPa]	69.7	95.4	4.94	5.03	4.93	4.76	4.91	5.05

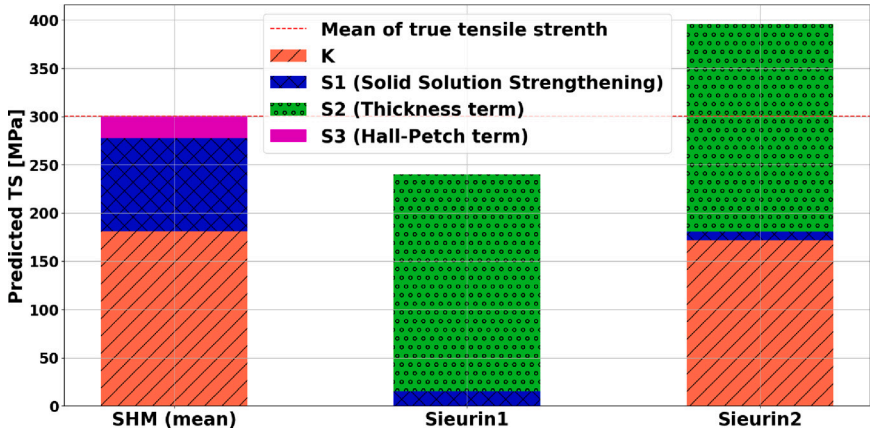


Fig. 1. Barchart of the mean contribution from the four terms in Eq. (3) to the prediction of the tensile strength (TS). The mean of the optimized parameters from fitting with five 5-fold cross-validation splits is shown as 'SHM (mean)' and the results with parameters taken from literature (Eq. 11 and Eq. 13 in Sieurin et al. (2006)) are denoted as Sieurin1 and Sieurin2, respectively.

4.2. AHM

The sub-expressions for the average grain size f_g found with GP using the structure template after re-fitting coefficients and manual simplification using the python sympy package (Meurer et al., 2017) are given by Eqs. (5) to (9). The process involves exporting the full AHM result as a python function in HL and manually selecting and simplifying f_g with sympy.simplify. The whole python functions of all 5 AHMs are provided in the supplementary information.

$$f_{g,0} = 0.12\ (\varepsilon\ (\varepsilon\ \exp\left(\frac{-10044.11\varepsilon + 1.09\ t_{\text{hold}} - 119384.02}{RT}\right) + 0.01\exp\left(\frac{-148570.90\ \varepsilon}{RT}\right))^{-0.5} \tag{5}$$

$$f_{g,1} = \left((27.74\ \varepsilon\exp\left(\frac{34151.02\ \varepsilon}{RT}\right) - 1107.91\varepsilon + 675.08)\exp\left(\frac{-40947.15\varepsilon + 1.21\ t_{\text{hold}} - 143110.53}{RT}\right)\right)^{-0.5} \tag{6}$$

$$f_{g,2} = 0.33\ \left((\varepsilon^4 - 0.75\ \varepsilon + 0.58)\exp\left(\frac{-24564.57\ \varepsilon + 1.19\ t_{\text{hold}} - 107031.29}{RT}\right)\right)^{0.5} \tag{7}$$

$$\begin{aligned} f_{g,3} = \Bigg(&- ((13.58\ \varepsilon - 36.14)\exp\left(\frac{10289.24\ \varepsilon}{RT}\right) \\ &+ (38.88\varepsilon - 27.46)(-(30.61\ \varepsilon - 21.43)\exp\left(\frac{-8946.93\ \varepsilon + 1.92\ t_{\text{hold}}}{RT}\right) + 1)) \\ &\exp\left(-\frac{10289.24\ \varepsilon + 123684.86}{RT}\right)\Bigg)^{-0.5} \end{aligned} \tag{8}$$

$$\begin{aligned} f_{g,4} = \Bigg(\Bigg(&50.61\ \epsilon\left(12.88\epsilon - \exp\left(\frac{14943.60\ \epsilon}{RT}\right) + 17.74\right) \\ &\exp\left(\frac{47422.20\ \epsilon}{RT}\right) + 36.17\ t_{\text{hold}}\Bigg)\exp\left(-\frac{47422.20\ \epsilon + 142088.24}{RT}\right)\Bigg)^{-0.5} \end{aligned} \tag{9}$$

The other coefficients of R_m according to Eq. (4) are listed in Table 4. The models for the average grain size f_g from Eqs. (5) to (9) achieved the best performance on the training set among 30 independent repetitions. MAE and R^2 are similar than the ones achieved by the SHM with a mean MAE of 4.921 MPa and a mean R^2 of 0.573, evaluated on the test set. The full list of results from the AHM is presented in Table 5. The activation energy terms Q_{rex} and Q_g are replaced by complicated terms dependent on ϵ and t_{hold} in the Arrhenius-like expressions within the exponential functions. While activation energies are material-dependent and considered fitting parameters in Eq. (3), ϵ and t_{hold} are process parameters independent of the material. The constant value of d_0 renders it inconsequential and therefore, it is not considered in any AHM.

Table 4
Comparison of the material dependent and fitting parameters in the five Analytical Hybrid Models (AHMs).

Parameter	AHM_0	AHM_1	AHM_2	AHM_3	AHM_4
A_{Al}	49.34	48.58	51.05	46.51	50.18
A_C	228.63	237.66	234.04	240.34	229.60
A_{Cr}	32.59	35.23	32.56	31.63	35.37
A_{Cu}	21.71	14.10	15.10	17.96	17.02
A_{Mn}	55.86	54.50	53.44	56.18	56.71
A_{Mo}	178.46	180.60	183.94	181.53	180.15
A_{Ni}	23.40	26.72	25.04	23.44	21.22
A_N	−535.67	−470.60	−542.21	−554.64	−429.49
A_P	311.78	312.87	310.11	308.29	305.36
A_{Si}	104.25	99.80	104.13	104.38	99.76
A_S	−18.78	−19.34	−12.87	−20.00	−20.93
A_{Sn}	−19.44	−9.78	−19.73	−14.88	−27.09
A_{Ti}	374.53	367.25	349.25	353.81	375.86
A_W	2.33	4.32	−0.83	−0.33	4.88
K	222.95	220.60	224.77	224.04	221.11
A_g	16.17	2.23	4.53	14.52	25.84
A_t	0.19	0.18	0.16	0.16	0.13

Table 5
Comparison of the prediction quality of our AHMs using a 5-fold cross-validation.

	AHM (Mean)	AHM_0	AHM_1	AHM_2	AHM_3	AHM_4
R2	0.573	0.561	0.579	0.597	0.565	0.561
MAE [MPA]	4.92	5.01	4.91	4.76	4.90	5.02

4.3. ABBM

Applying SR without restrictions allows for more complex analytic relations. One result of GP-based SR without structure template is given by

$$
\begin{aligned}
\mathrm{ABBM}_2 = &\left(\frac{T - 3.82\times10^{17} + c_P^{2/3}2.78\times10^{19} + c_{Si}^{2/3} - 2.08\times10^{20}}{c_{Al}^{2/3} - 2.89\times10^{18} - (c_N^{1/2}1.84\times10^{18} + c_C^{2/3} - 3.92\times10^{17} + 1.70\times10^{17})}\right.\\
&\left. + \frac{(T - 9.25\times10^{11} - 7.57\times10^{13})c_N^{1/2} - 3679063.6 - 2.06\times10^{20}}{c_{Al}^{2/3} - 2.89\times10^{18} - (c_N^{1/2}1.84\times10^{18} + c_C^{2/3} - 3.92\times10^{17} + 1.70\times10^{17})}\right)\\
&+ \left(\frac{(c_C^{2/3} - 0.140 + T - 0.00062 + c_{Ni}^{2/3}0.0125 + c_N^{1/2})T - 5.47\times10^{12}}{c_{Al}^{2/3} - 2.85\times10^{10} - (T6.25\times10^{7} + c_N^{1/2}1.32\times10^{10} - 2.39\times10^{9})}\right)\\
&+ \left(\frac{\epsilon\ 1.23}{c_N^{1/2}} + \frac{0.738}{c_N^{1/2}}\right)1.003\\
&+ c_P^{2/3^2}2360.701 + c_P^{2/3}c_{Ti}^{2/3}9146.251 + \sqrt{T} - 380.734\\
&+ c_{Cr}^{2/3}36.785 + c_{Mn}^{2/3}49.173 + c_S^{2/3} - 14.408\\
&+ c_P^{2/3}84.294 + c_{Mo}^{2/3}180.367 + \epsilon - 20.682\\
&+ T59.954 + c_{Ni}^{2/3} - 1029.354 + c_N^{1/2} - 86203.557\\
&+ c_{Si}^{2/3} - 227.489 + c_{Al}^{2/3} - 18494.696 + c_C^{2/3}11697.673 + 6962.479
\end{aligned}
\tag{10}
$$

This model achieved the best performance on the training set among 30 independent repetitions for the second cross-validation fold. Equations of the best models for the other four cross-validation folds are provided in the supplementary information and the resulting metrics are shown in Table 6. The ABBM approach achieves a better MAE of 4.652 and R^2 of 0.608 on the test set compared to the AHMs, but has more fitting parameters, more complexity, and no physical meaning that can be derived from it due to its black box nature. Eq. (10) does not use the operators exp and log, although they are included in the GP function set. Also not all features are used in $ABBM_2$, pointing to the fact, that not all features contribute significant information to the target parameters. These features include the Cu, Sn, and W content, as well as the holding time during annealing and the thickness. Later, we will see that those features are attributed limited significance also in the ANN. However, equations from folds 0, 1, 3, and 4 do use some of these operators and features, while leaving out others, as shown in the supplementary information. The Cu and Sn content is absent in all of the algebraic expressions.

Table 6
Comparison of the prediction quality of our ABBMs using a 5-fold cross-validation.

	ABBM (mean)	ABBM_0	ABBM_1	ABBM_2	ABBM_3	ABBM_4
R^2	0.591	0.573	0.594	0.608	0.589	0.589
MAE [MPA]	4.81	4.95	4.85	4.65	4.74	4.86

Table 7
Resulting hyperparameters and prediction quality of our ANN with a 5-fold cross-validation including an automated hyperparameter search. λ_{L2} corresponds to the L2 regularization strength.

	ANN0	ANN1	ANN2	ANN3	ANN4
Batch size	996	669	421	608	510
Learning rate (start)	0.000278	0.000567	0.002055	0.006333	0.002484
Learning rate factor	0.000939	0.556511	0.290281	0.189539	0.79051
Number of hidden layers	3	10	9	8	3
Number of nodes per hidden layer	162	144	180	180	90
λ_{L2}	0.00076	0.00117	0.002949	0.00107	0.00073
R^2	0.620	0.660	0.668	0.667	0.669
MAE (MPa)	4.60	4.36	4.24	4.29	4.37

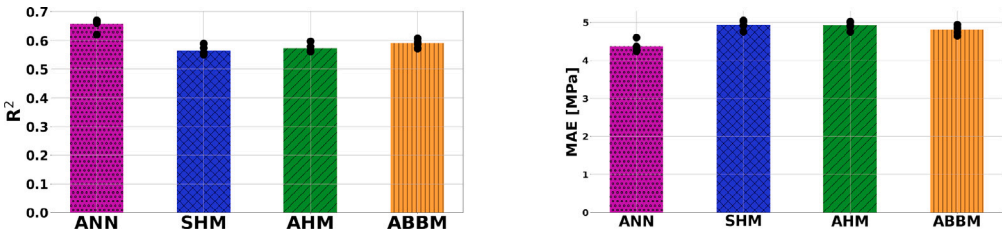


Fig. 2. Barplots comparing the performance of artificial neural networks (ANN), Sieurin–Hodgson model (SHM), the analytical hybrid models (AHM), and the analytical black-box models (ABBM). The height of the bar corresponds to the mean value of R^2 and MAE in a 5-fold cross validation, while the black points are indicating the individual results.

4.4. ANN

We trained artificial neural networks with an automated hyperparameter determination, as described in the previous chapter, with the same dataset used for the other experiments. With a mean R^2 of 0.66 and a mean MAE of 4.37 MPa in a 5-fold cross-validation, ANN outperformed SHM, AHM, and ABBM, see Table 7. A comparison of results from the four modeling methods is shown in Fig. 2.

4.5. Feature importance determination and impact of chemistry

In this section we investigate the impact of chemical elements on the prediction of tensile strength in all our models and compare the captured impacts. In the case of our SH models, AHMs and ABBM, we can directly determine the effect from the optimized parameters. A similar analysis is also possible for the employed ANN despite its general consideration as black-box model using SHAP (SHapley Additive exPlanations) (Lundberg & Lee, 2017). Employing SHAP enables the determination of a feature's impact on a (black-box) model by adjusting the model input and calculating contributions across all possible permutations. In short, SHAP values for a feature reveal how much the feature alters the predicted value in the corresponding model, starting from the mean prediction value for each specific observation in the dataset. For a more comprehensive insight, see Molnar (2022, 2023).

We calculated SHAP values for our ANN and show the beeswarm plot in Fig. 3. The coloring of the SHAP values in the beeswarm plot is according to the feature value, therefore, a clear linear trend can be seen by a continuous color change.

In literature (Duan et al., 2022; Hudd, 2006; Hutchinson, 1984; Miranda et al., 2019; Sieurin et al., 2006), the relationship between chemical elements and tensile strength is often described solely with a linear solid solution strengthening effect, usually with a positive contribution. This suggests a monotonous linear rising trend, also shown by most of the elements in our models. In our optimized SH models, only S and Sn have a (small) negative prefactor. Similar negative prefactors for S and Sn are obtained with our AHM, which in addition yield a larger negative factor for N as well. In the case of the ABBM (Eq. (10)), the contributions are often split up in multiple terms, allowing non-linear impact of chemistry. The variable impact was determined by calculating the SHAP values and the beeswarm plot in Fig. 3 shows a positive correlation of tensile strength and nearly all chemical elements, except Al, where the correlation is non-linear. Regarding our ANN, the impact is clearly ascending for N (shown in Fig. 4), C, P, Si, Mo, Ni, Mn, Al, and Cr, but the trend is less obvious for Cu (shown in Fig. 4), S, Ti, W, and Sn. It is interesting to note that Cu and Sn are not even considered in any ABBM, pointing to their low significance.

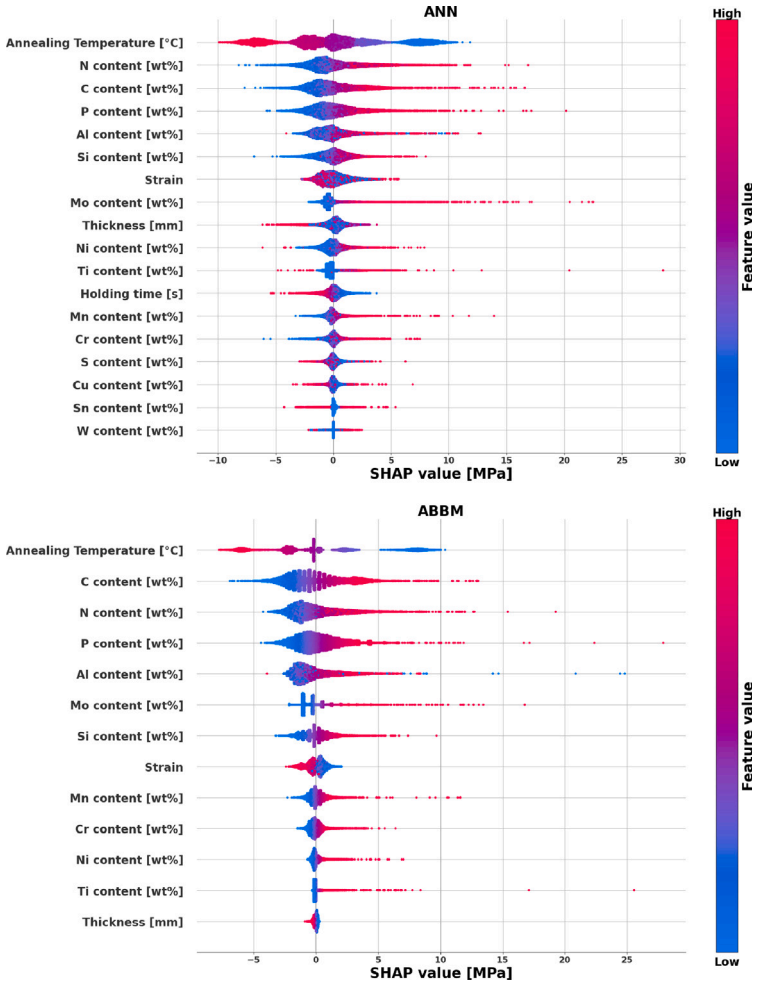


Fig. 3. Beeswarm plots of ANN (trained on the complete data set) and the $ABBM_2$, Eq. (10).

The SHAP values of Al are showing a slightly non-linear dependence, as can be seen in Fig. 5 in the ANN and the ABBM, contradicting the prevailing opinion in literature (Duan et al., 2022; Hudd, 2006; Hutchinson, 1984; Miranda et al., 2019; Sieurin et al., 2006). While previous studies have documented a non-linear relationship in bcc materials, such as of Re in Mo (Ref. Trinkle and Woodward (2005)), the authors are not aware of such a behavior in the case of Al within ferrite. The course initially descends, then ascends again once it reaches a minima. Although the visible effect occurs in very few points with low Al-content, exploring it further is considered worthwhile. The high SHAP values for relatively low Al content can be caused by the fact that SHAP values show the whole contribution of Al in the ANN, not only the solute solution strengthening part. Therefore, other effects e.g. arising due to the precipitation of AlN particles are overlaying the linear strengthening effect as Al removes the nitrogen in solution at low additions (Mintz, Gunawardana, & Su, 2008; Mintz, Williamson, Su, & Morrison, 2007).

The top five chemical elements with the highest impact in our ANN are N, C, P, Al, and Si according to the mean absolute SHAP value (see Fig. 3). While the five chemical elements with the highest mean absolute SHAP value in our ABBM are C, N, P, Al, and Mo. Regarding the SH model, Ti, N, P, C, and Mo hold the highest fitting parameters, exerting notable influence, whereas in the AHM, N, Ti, P, C, Si, and Mo exhibit the greatest absolute parameter values. The significant impact of N, C, and P consistently emerges across all models, underlining their important role in solute solution strengthening. This fits to what one would expect according to literature. E.g. Pickering and Gladman (Steels, 1963) explored the influence of alloying elements on the tensile strength of ferrite grains in and identified C, N, and P as the elements with the strongest impact. C and N are known to occupy interstitial sites in the Fe lattice, due to their small size, causing a huge strengthening effect Sieurin et al. (2006) and Gladman (2002). Despite comprehensive literature research, we encountered difficulty to pin down the causative factor underlying the pronounced prominence of P relative to other comparable elements. As commented by Gladman (2002), the explanation remains unclear, potentially tied to chemical, electrical, or geometrical interactions. One possible reason might be strong electrochemical effects of P and its role as compound former, leading to interatomic forces boosting the strengthening effect Steels (1963).

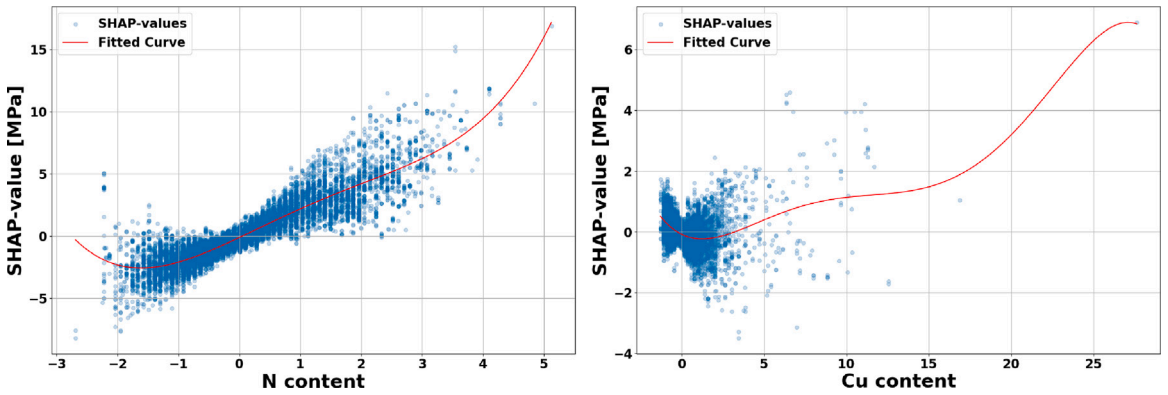


Fig. 4. SHAP values of the N and Cu content of our ANN. The values of the features are normalized with Z-score normalization, darker areas show higher point density and a polynomial fit of 5th order is shown in red.

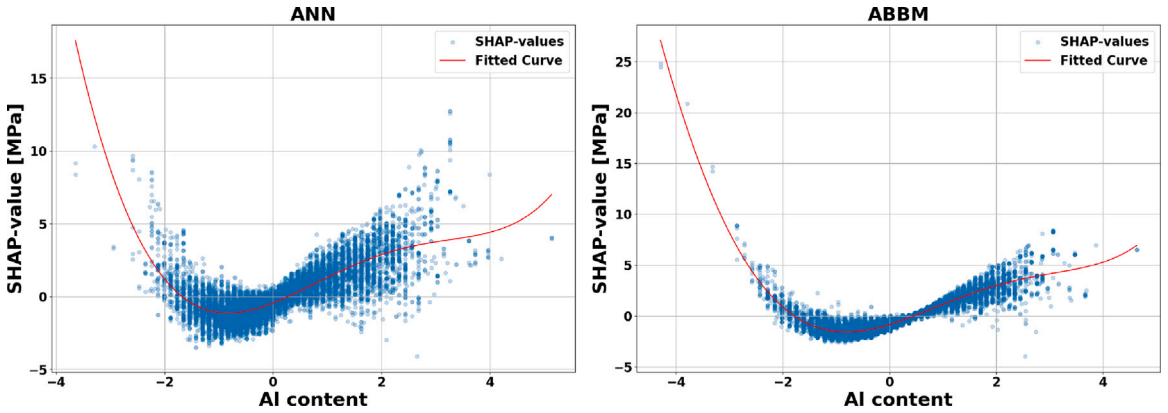


Fig. 5. SHAP values of Al content of our ANN (left) and the ABBM_2 (right), both showing a non-linear relationship. The values of the features are normalized with Z-score normalization, darker areas show higher point density and a polynomial fit of 5th order is shown in red.

5. Conclusion

In this work we presented data-driven models to predict tensile strength of cold-rolled, batch-annealed low-carbon steel coils from mostly compositional information. We compared the prediction quality of a semi-empiric model derived from literature, Analytical strengthening models derived with the help of symbolic regression with and without partial fixing of the structure, due to physical considerations, and ANN as a classic machine learning model. Additionally, we investigated the influence of chemical elements on the solute solution strengthening effect in all our models. The following findings were made:

(i) Our ANNs have the best prediction quality of all investigated models, with a 5-fold cross-validation R^2 of 0.66 and a MAE of 4.37 MPa. Although the predictions are not as easily traceable as those from the developed SHMs and AHMs, we demonstrate that interpretability persists through the calculation of SHAP values.
(ii) We were able to find similar algebraic expression for the grain size evolution as described by Hodgson and Gibbs (1992) via genetic programming for SR and physics-inspired structure constraints with a comparable prediction quality.
(iii) Nearly all considered chemical elements (Al, C, Cr, Cu, Mn, Mo, N, Ni, P, S, Si, Sn, Ti, W) show a positive influence in all investigated models. The strongest impacts are found for C, N, and P, which is in accordance with literature sources.
(iv) The SHAP values concerning Al reveal a nonlinear trend, indicating a significant negative impact at lower Al contents. This behavior may be attributed to other factors masking the solid solution strengthening of Al. To further investigate this impact of Al, datasets with systematic variations of Al and N contents could be of help.

While our results are encouraging, limitations exist. Our models mostly use compositional data with few processing parameters; expanding it with more process parameters like detailed information about deformation forces would improve the prediction quality, as also shown in previous works (Millner et al., 2023, 2024). Future work should also incorporate additional microstructural features (e.g., dislocation density, precipitates, texture) and expand to other mechanical properties (e.g., yield strength, elongation, plastic strain ratio).

CRediT authorship contribution statement

Gerfried Millner: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Data curation. **Gabriel Kronberger:** Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Manfred Mücke:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization. **Lorenz Romaner:** Writing – review & editing, Supervision, Formal analysis, Conceptualization. **Daniel Scheiber:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT (3.5 and 4.0) and Claude 3.5 in order to check grammar and improve readability. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary information on the results of AHM and ABBM models is provided in a PDF file. Along with supplementary equations of all ABBM models, beeswarm plots of the ABBM models and dependence plots of ABBM models regarding Al are provided. Additionally, python equations of all AHM and ABBM models are provided in the online version to make a re-use more convenient.

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijengsci.2025.104247.

Data availability

The authors do not have permission to share data.

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